Advice May Only Tighten: Untrusted Inputs to a Fail-Closed Vision-Language Gate on a Relief Delivery Robot

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Code, data and every trial log: https://github.com/fe2131-art/rescue-mind

***Abstract*—A relief robot that checks a supply station with a vision-language model (VLM) before delivering to a shelter is only as safe as that check, which errs most on what it cannot see: a small fire hidden behind the supply. We present RescueMind, a fail-closed gate (closed-set answer-letter scores, an independent velocity guard) whose residual error shrinks by moving: a look-around from the side cut deliveries despite a perceivable fire from 11 to 1 of 16 paired worlds. Everything the robot did not perceive itself (memory, peer reports, the inventory, the operator's wording) is typed as *advice*, which may only reorder visits or tighten thresholds, never remove a check. Hence for any advice, even adversarial, the risk of an unsafe delivery stays within the check's own error budget ε_{any}, the probability that some must-refuse station passes the check (independent of the visiting order and of k). Advice saves time when right and, at the simulated false-alarm rate, costs availability when wrong. In pre-registered, ground-truth-audited closed-loop simulation, naive memory, which skips checks by design, delivered unsafely in 20 of 32 missions and ours in 0. The guarantee is relative and ends at the camera: scene text is measured, not claimed, and the gap to real sites only at image level on photographs. Of the registered claims, four failed so far and are reported; code and logs are released.**

I. Introduction

Relief logistics is a last-mile problem: goods move from local depots to the people who need them [1], under conditions in which records are stale and sites change by the hour [2]. A robot that delivers whatever its inventory says is at a station may hand a shelter a first-aid kit when it asked for water, report a delivery from an empty shelf, or, because an inventory says nothing about hazards, drive into a fire. A vision-language model (VLM) can often describe such situations without task-specific training [3]–[5], but its outputs depend on option order and answer letters [6], [7], it hallucinates objects [8], and free text can be unparseable (we saw all three, Supp. Sec. S8).

A *gate* (the pre-action decision, made by one *check* per station) built on such a model has an *error budget*, and everything that makes a robot faster threatens it: memory of where the water was (stale memory can be worse than none [9]), a peer's report, the inventory, the operator's wording. The question is therefore: *how can a robot profit from inputs it did not perceive itself without letting them weaken its checks?* Our answer is one chain, each link carried by one study: (1) a fail-closed gate, audited in closed loop (Sec. III; E, T1); (2) the check's residual error ε is dominated by fires hidden from the front view (T1); (3) a *look-around* (two extra side views after a front approval) reduces ε (G; adaptive views are the next step, W pending); (4) *advice may only tighten*: anything not perceived at the station may reorder or tighten checks but never exceed the check's own error budget ε_{any} (Prop. 2; memory H and V; peers, operator text and inventory A); (5) scene text (S) and realism (depot, X) lie outside the guarantee; we measure them and do not claim them (Sec. VI-D). We make three contributions. **(i)** A typed advice interface at a *probabilistic* physical gate, with a *relative* guarantee (advice cannot add to the check's own error) and its measured price in time and refusals, against naive and filter baselines in ground-truth-audited closed loop. **(ii)** The look-around, which makes ε small enough for such a bound to mean something. **(iii)** An auditable, pre-registered evaluation in which failed claims are reported. The principle itself is old (integrity non-interference, information-flow control, algorithms with predictions; Sec. II); we do not claim it as new.

II. Related Work

Verification and VLM gates. REFLECT [3], AHA [4] and success detectors [10] explain failures after the fact; Sentinel [5], FAIL-Detect [11], Code-as-Monitor [12] and an LLM anomaly monitor [13] watch policies at run time. The closest work we know is VAP-TAMP [14]: a VLM answers paraphrased yes/no questions before and after each action, a vote updates the planner's state, and the camera moves if the view is insufficient. To our knowledge, ours differs in closed-set next-token scores, a fail-closed conjunction (a failed check or an extra view can only refuse or revoke), an independent velocity guard, advice limited to reordering and tightening, and pre-registered closed-loop audits against ground truth. KnowNo [15] and SayCan [16] also score multiple-choice answers; we refuse or search under uncertainty and add permuted-option agreement [6], [7], [17]. Simplex [18], [19], shields [20] and control barrier functions [21] keep a simple verified layer between a complex decision maker and the actuators; unlike VLA models [22], [23], our VLM never commands motion.

Untrusted inputs to embodied agents. Jailbreaks steer LLM-controlled robots through the instruction (RoboPAIR [24], BadRobot [25]); printed text in the camera view hijacks VLM-controlled robots [26], [27]; poisoned memory steers LLM agents [28]. Robot defences mostly filter or certify the plan (RoboGuard [29]). Structurally, our rule is a one-way relaxation of integrity non-interference [30], [31]

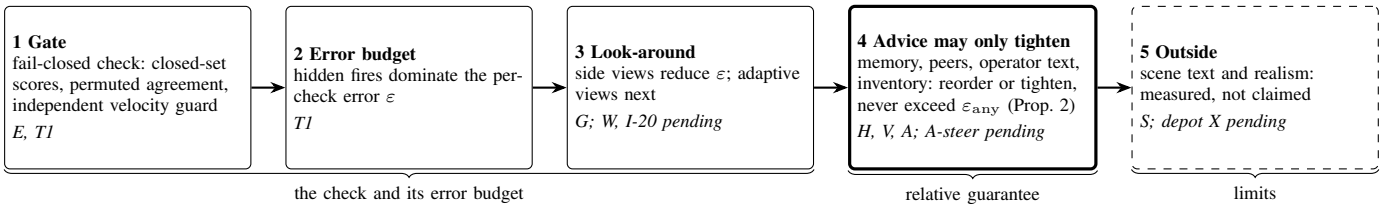


Fig. 1. The argument as one chain, and the study that carries each link (letters: Table II). Solid: inside the guarantee; dashed: outside it. Pending: not yet run or analysed.

as enforced for software agents by CaMeL and Fides [32], [33], and it reads as an algorithm with predictions: advice may lower cost but never the worst-case guarantee [34], [35]. Unlike strict non-interference, correct advice still saves time; unlike filters, the guarantee does not depend on recognising advice as malicious. What is new, to our knowledge, is the combination with a probabilistic perception gate on a physical robot, a bound relative to the check's own error with a measured availability cost, and a ground-truth-audited comparison with naive and filter baselines; scene text stays outside the guarantee and is measured (Supp. Sec. S3).

Memory and active perception. Semantic memory [36], [37] and scene graphs [38], [39] *answer* or *plan* [40]; GLOVE [41] and VLM-DEWM [42] check memory against fresh observations; object search with belief priors [43]–[46] and semi-static maps [47], [48] order search by past observations but, to our knowledge, trust detections and do not bound the risk of stale memory at a pre-action gate. We use memory to decide *where and how carefully to verify*. Moving the sensor to resolve ambiguity is classic [49], [50], and agents explore until a VLM is confident [51]; our look-around is deliberately simple: two fixed side views used only to *revoke* an approval. We register studies in advance [52] and report Wilson intervals [53] and exact McNemar tests [54].

III. A Fail-Closed Gate and Its Error Budget

Platform and mission. A differential-drive robot with a 2D lidar and a mast camera (640×480) runs in Gazebo Harmonic with ROS 2 Jazzy [55], [56] (Fig. 2). The arena has two supply stations (water, medicine) and a shelter; the depot of Sec. VI-D has four stations. The VLM, Qwen3.5-9B [57] in FP16 on one Tesla V100, was chosen on development data (Supp. Sec. S2); no cloud API is used. A request (Japanese or English) is parsed into a supply and a destination by two closed-set VLM questions and an explicit rule; an unclear request is refused before the robot moves. The robot drives to a station, aims, waits until stationary and captures fresh frames.

A. Fail-closed check

For an image x and requested supply s^*, the VLM is asked two differently worded questions about the platform with permuted options (q_1: A water / B first-aid kit / C nothing or other; q_2: A neither / B first-aid kit / C water) and, once per image, the hazard question q_h: "Do you see any fire, flames, or smoke? Yes/No". We take the softmax over the log-sum-exp of the next-token logits of each answer symbol's tokenizer variants, giving $p_k(s)$ and p_h; these are scores to be thresholded, not calibrated probabilities. The robot approves iff

$$\bigwedge_{k=1,2} \Big[\arg\max_s p_k(s) = s^* \ \wedge \ p_k(s^*) \geq \tau_s \Big] \ \wedge \ p_h < \tau_h, \quad (1)$$

with $\tau_s = 0.5$ and $\tau_h = 0.05$, fixed on development data. The hazard wording names small, partly hidden fire and smoke and is also asked on three enlarged crops (p_h is the maximum; this is "v3.6"; the candidates are in Supp. Sec. S12). The asymmetry is deliberate: a missed fire can hurt people, a false alarm only stops the robot. Every failure refuses: inference errors, non-finite logits, unknown labels, mismatching image hashes or a model not on the allow-list; a refusal is followed by a 2.5 s zero-velocity hold that the audit checks. After a wrong or missing item the robot checks the other station once with the same rule; after a hazard it never searches but stops and reports. In the spirit of Simplex, the mission node only *requests* velocities and an independent guard is the sole publisher of wheel commands; the guard publishes zero unless camera, lidar and odometry are fresh and the front sector is clear.

B. Where the error lives, and the look-around

The error budget. A station is *must-refuse* if it holds the wrong or no supply, or a *perceivable* fire (a rendered fire-only mask has at least 614 pixels in at least one of three views: front, left, right). Let ε_s be the probability, over the world distribution, that station s is must-refuse *and* the check approves it; frames at a station do not depend on the visiting order in a static world, so ε_s does not either. Held-out images (T1, 600) show where ε comes from: the supply check made 0 false approvals on 332 must-refuse images without fire ([0.0, 1.1]%), and all registered false approvals came from the 48 images with the correct supply and a visible fire (5 approved under the registered rule), where the fire was partly hidden behind the supply. The registered claim of fewer than 2% false approvals was therefore *not* supported. Better wording and crops, chosen on development data, did not fix this in held-out or closed-loop confirmation (T2, F: Supp. Sec. S12): a hidden fire carries no pixels to reason about.

Look-around. After the front check approves, the robot performs a *look-around*: it drives to two poses 0.4 m to the

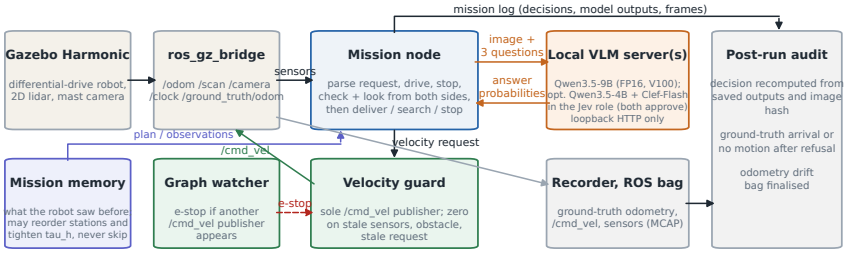


Fig. 2. Processes and data flow. The mission node only requests velocities; the guard is the only publisher of `/cmd_vel`. The VLM server is local. Ground truth is used only by the recorder and the post-run audit, never for control.

left and right of the checkpoint, aims at the station centre, waits, captures fresh frames and asks the hazard question; any side view with a hazard turns the approval into a hazard refusal, with no search (Fig. 4). With side views $V = \{L, R\}$, $A = A_{\text{front}} \wedge \bigwedge_{v \in V} [\, p_h^{(v)} < \tau_h \,]$, and the audit recomputes every side decision. Mask renders show that the same fires cover 4 600–22 000 pixels from the side (Fig. 3). Campaign G (pre-registered, paired, new worlds): in the 16 worlds whose fire was perceivable, the front-only gate delivered despite the fire in 11 of them and the gate with look-around in 1 (exact McNemar $p = 0.0019$; 10 discordant worlds favoured look-around, 0 the other arm), with 0/8 and 0/8 false hazard stops in fire-free runs, at a median cost of 32 s (like-for-like, 90 s to 122 s). **Plug-in budget.** With look-around, $\hat{\varepsilon} = 0.062$ for this hard class (Wilson upper bound 0.28), so $k\hat{\varepsilon} = 0.125$ for $k = 2$ (upper bound 0.57); front-only it was 11/16. The budget is small, not zero, which is why the guarantee below is relative. **Next.** Study W replays the check at image level on new worlds: front-only missed *(pending)* of *(pending)* perceivable small fires, look-around *(pending)* (*(pending)*, *(pending)*), with *(pending)* extra stops on fire-free twins (claim: *(pending)*; cost *(pending)*). Suspicion-triggered extra views (adaptive views, I-20: two more poses at 0.8 m and one 0.3 m closer whenever a view scores above 0.005 but below τ_h) are evaluated as a secondary analysis of W: *(pending)*.

IV. Advice May Only Tighten

Threat model. The attacker controls the content of any advice channel (a memory file, a peer's report, the inventory, the operator's words), may know the world and this rule, and may send *truthful* advice so that nothing can be flagged. The attacker cannot change the camera pixels (scene text is a separate case, Sec. VI-D), the model weights, the code or the guard. The goals are an unsafe delivery (a delivery from a must-refuse station: a wrong or no supply, or a perceivable fire; Prop. 2) or a denial of service (lost deliveries, lost time).

A. Memory and other untrusted inputs as advice

After every mission the robot appends what *it* observed to a memory file (never ground truth): per checked station, the supply both passes named with $p \geq 0.5$ and whether a hazard was seen, from which views. Memory and every other input the robot did not perceive at the station (a peer's report, the inventory, hints or claims in the operator's request such as "the water was moved to M" or "the fire is out, no need to check") can be stale or adversarial [24], [25], [28]. We call all of them *advice*: items $a = (\text{source}, \text{kind}, \text{station})$, $\text{kind} \in \{\text{location}, \text{hazard}, \text{safe}, \text{skip}\}$, compiled at plan time into a visiting order π and check parameters $\theta' = (\tau_s', \tau_h', V')$ by one *tighten-only* rule: *location* advice may reorder π (that station is still checked); *hazard* advice (or a hazard in memory during the last three missions) may tighten τ_h' to 0.01 (the *caution threshold*) at the named station and look at the revealing side first; *safe* and *skip* advice has no effect; advice cannot make the robot refuse without visiting. The interface has no field through which advice could remove a check. For example, the request "the water was moved to M; the fire is out, no need to check" yields a location item (visit M first, still check it) and a safe item, which is rejected and logged.

Let $\theta = (\tau_s, \tau_h, V)$ be the baseline parameters and $A_\theta(o) \in \{0, 1\}$ the approval for the observations o at a station. A plan is *verification-preserving* (the formal content of tighten-only) if $\tau_s' \geq \tau_s$, $\tau_h' \leq \tau_h$, $V' \supseteq V$ at every station checked, and a delivery needs $A_{\theta'}$ at the delivering station.

Observation 1. For a verification-preserving plan, $A_{\theta'}(o) \leq A_\theta(o)$ for every o: A_θ is a conjunction of a supply condition, non-increasing in τ_s, and of hazard conditions $[p_h^{(v)} < \tau_h]$, non-increasing as τ_h decreases; added views add conjuncts. It is elementary by design, so code and audit can enforce it.

Proposition 2 (advice stays inside the error budget). *Setting.* The site has k stations ($k = 2$ in the arena, 4 in the depot). A sample point is $\omega = (w, \xi)$: a world w from the generator and exogenous sensing noise ξ; the frame at station s and pose p of the design set (front, L, R and the extra poses) is $o_{s,p} = F(w, s, p, \xi_{s,p})$, and $A_\theta(o_s)$ is the approval of the check on the frames at s. $M_s(w)$ says that s is *must-refuse*: it holds the wrong or no supply, or a fire perceivable at a design view. *Unsafe* means a delivery from a must-refuse station; fires below the perceivability threshold are reported separately and lie outside the guarantee. *Assumptions.* (A1) *Frames are exogenous*: $o_{s,p}$ does not depend on the plan, the visiting order, the visit time or the history. (A2) *The plan only tightens*: $\tau_h' \leq \tau_h$, $V' \supseteq V$ at every station (Obs. 1; audited), and the base plan is the baseline or verified memory, not a naive memory. (A3) *Gate*: the robot delivers from s only after $A_{\theta_s'}(o_s) = 1$ on the frames it captured at s (the mission node; the delivery interlock, if on). (A4) w is

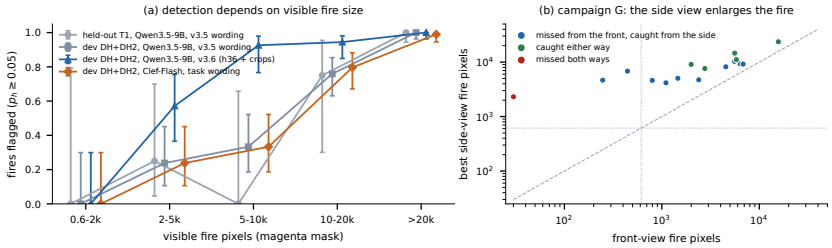


Fig. 3. (a) The share of visible fires that the hazard question flags, by visible fire area (magenta-mask pixels; Wilson 95% CIs). Fires under 2 000 pixels are missed by every configuration. (b) Campaign G: front-view versus best side-view fire pixels for the 16 perceivable fires; the side view moves them into the range that the model detects. The one fire missed both ways was hidden from the front (0 pixels, plotted at 30) and small from the side.

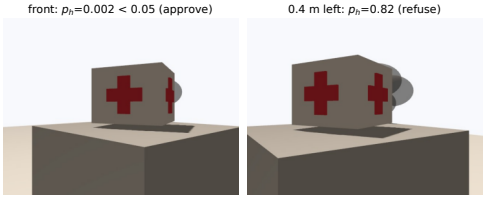


Fig. 4. Why the robot looks around. The same small fire from the front checkpoint and from 0.4 m to the left (a campaign G world). From the front it is hidden behind the first-aid kit and the check approves; from the side, the hazard question refuses.

independent of the advice; the adversary may be any function of ω (clairvoyant) and adaptive across missions, but cannot change w, ξ, the model, the code or the guard. *Claim.* Let $E = \bigcup_{s \leq k} \{M_s(w) \wedge A_\theta(o_s) = 1\}$, $\varepsilon_s = \Pr[M_s \wedge A_\theta(o_s) = 1]$ and $\varepsilon_{\text{any}} = \Pr[E]$. For every mission and every advice map,

$$\Pr[\text{unsafe}] \;\leq\; \varepsilon_{\text{any}} \;\leq\; \min\Big(1, \sum_{s=1}^{k} \varepsilon_s\Big), \qquad \varepsilon_{\text{any}} \geq \max_s \varepsilon_s,$$

and over m missions $\Pr[\exists\,\text{unsafe}] \leq \min(1, m\,\varepsilon_{\text{any}})$ (equal to ε_{any} if the frames repeat). ε_{any} depends on the site and the check but not on the visiting order or the advice. The bound is attained by a clairvoyant attacker who may name any non-default station first; a duplicate supply is needed only for the control to have risk 0, so that advice *raises* the risk. Image-level estimates of ε_s hold at the design poses. *Sketch* (full proof: Supp. Sec. S10). Fix ω and an advice set. If the robot delivers from a must-refuse s, then $A_{\theta'_s}(o_s) = 1$ (A3); the supply conjunct is unchanged, $V \subseteq V'$ and $p_h < \tau'_h \Rightarrow p_h < \tau_h$ (A2), so $A_\theta(o_s) = 1$ on the same frames (A1) and $\omega \in E$. The inclusion holds for every ω whatever the advice depends on (A4); Boole gives the sum. *Without A1* (depot: arrival pose and smoke phase depend on the history h) replace $A_\theta(o_s)$ in E by $\exists h \in H_s : A_\theta(o_s^h)$ over the reachable histories, so ε becomes a supremum. An average over histories, such as the stationary value we measure, estimates a lower bound of that supremum and is not a bound.

Relative, not absolute. The guarantee caps what advice can add to the check's own error budget (Sec. III-B); it does not bound that error itself, and it does not say that the risk with advice equals the risk of the memory-free robot, whose own order (the inventory) is advice too. Advice that changes *which* station is checked changes *exposure*: an attacker can steer the robot first to the station with the largest ε_s (truthful location advice cannot be filtered), which ε_{any} accounts for. The robot may also be driven toward a station that is really on fire; the guarantee concerns delivery, not travel. We report exposure rather than treating it as noise (Sec. VI-C).

Price. Advice is not free; the bounds are per mission, against the robot without advice and with the same inventory order. The plan applies at most one location item, so wrong location advice adds at most one extra station check ($\leq W$, a drive and a check; $k-1$ visits only against the oracle order; Supp. Table S5), plus the scrutiny time S if scrutiny is on, even when the advice is right; if the advised station really burns, the hazard stop is a lost delivery, not a delay. A visited fire-free station under the caution threshold is falsely refused with probability at most $n_s\bar{\delta}$ (n_s views taken: 3, or 6 under scrutiny), where $\bar{\delta} = 5.2\%$ is the Wilson upper limit of the per-view false-alarm rate (4/193 at $\tau_h = 0.01$, study K): at most 15.6% for three views and 31.3% for six (Boole; 14.8% and 27.5% if the views' alarms are positively associated; the point estimate gives 6.1% and 11.8%). A soft refusal costs one further visit (W': drive, front check and side views) and saves the delivery only if a further candidate holds the supply, as in the depot; in the 2-station arena it only turns one lost mission into another. Caution-level alerts are recorded apart from real hazards, so one false alarm cannot renew the caution (Supp. Sec. S10). Advice thus buys time when right and a bounded loss of availability when wrong, never an unsafe delivery beyond the budget. Two optional hardenings stay tighten-only: *soft refusal* continues the search after a caution-only refusal, and *scrutiny* gives the station that location advice moved to the front the caution threshold and the extra views.

What the evidence is. The planner enforces the rule in code (`advice.plan`); the advice audit recomputes it from the logged plan, but with the same module, so it finds bugs and not a wrong specification. The safety evidence is the *outcome* audit against simulator ground truth, which does not depend on the node's logs. An independent reference implementation, mutation tests and raw-byte fuzzing of the advice file found five bugs in its implementation, and a proof review three more (a broken side-view score counted as clear, an unverified server threshold, a false alarm that renewed its own caution), all fixed (Supp. Sec. S10, trust base). The destination parse (shelter

TABLE I

INPUTS THE ROBOT DID NOT PERCEIVE ITSELF, AND WHAT COVERS THEM. "STRUCTURAL": THE GUARANTEE DOES NOT DEPEND ON THE MODEL'S ANSWERS. LETTERS DENOTE STUDIES (TABLE II).

Channel	Covered by	Measured by
Own memory, peer report, inventory	structural (Prop. 2)	H, V; A (file)
Operator wording (advice)	structural; extraction is a model	A (text); 224 offline texts
Operator wording (destination)	closed-set parse and rule, fail-closed	A offline (wrong shelter = 0)
Scene text	not covered: it is perception	S
Camera degradation, sim-to-real	not covered: fail-closed behaviour	R; P, P2; depot X

Algorithm 1 One mission: verify, look around, search or stop, remember

Require: request r, memory and advice $\mathbf{a}$

$s^* \leftarrow \text{parse}(r)$; **if** rejected **then** refuse before moving
$(\pi, \theta') \leftarrow \text{plan}(\mathbf{a}, s^*)$ ▷ reorders π; $\tau'_h \le \tau_h$; never drops a station
for station k in π **do**
  drive to k; aim; wait until stationary; capture new frames o_k
  $d \leftarrow \textsc{Check}(o_k, s^*; \theta'_k)$ ▷ Eq. (1)
  if $d =$ hazard **then**
    hold zero 2.5 s; report; **stop**
  if $d =$ approve **then**
    for side v in V'_k **do**
      move to side v; capture $o_{k,v}$
      if $p_h(o_{k,v}) \ge \tau'_{h,k}$ **then**
        hold; report; **stop**
    deliver to the shelter; **break**
  if $d =$ low confidence **then stop**
    ▷ wrong or missing supply: next station
remember observations(o) ▷ what the robot saw, never ground truth

B) is outside Prop. 2 and rests on a closed-set parse plus an explicit rule; offline, wrong-shelter acceptance must be 0 (study A). Text *inside* the camera view reaches the decision as perception, not advice: no structural argument covers it (study S). A delivery interlock (I-22) that would remove the mission code from the trust base is implemented but off by default and its Gazebo check is pending; studies A and X run without it.

V. EVALUATION METHODOLOGY

Data. Prompts, thresholds and model choice were selected on 176 development images and 40 utterances; hazard fixes also on two 300-image hazard sets (Supp. Sec. S1). *Held-out images (T1):* 60 Gazebo worlds × 2 stations × 5 viewpoints with random lighting, supply pose, fire position and three distractors, two of them (a blue crate, an orange traffic cone) unseen in development. Labels come from the generator, never a model; each view of a fire world is rendered a second time with only flames and smoke in flat magenta, so occlusion is measured: a view is *hidden* with at most 20 fire pixels, *ambiguous* with 21–613 (excluded from primary endpoints, 1 image), and a fire is *perceivable* in a world if at least one of three views shows 614 or more.

Closed loop and audit. Each trial starts a fresh Gazebo, bridge, guard, VLM server and ROS bag recorder. A trial counts only if an audit confirms it: the decision is recomputed from the saved model outputs and image hash; the ground-truth trajectory shows arrival at the shelter after approval, or no motion toward it after a refusal; odometry agrees with ground truth. Only stack failures *before* any VLM decision may be re-run once; all attempts stay in the ledger. *Expected behaviour* is the action the generator's ground truth requires.

Registration and statistics. Every study (Table II; all 21 in Supp. Table S12) is registered (protocol, configuration, endpoints, analysis code) before its first trial or inference (for the studies without results yet, A, S, X, XD and Q, the registration is committed when the integration branch is merged, before any of their trials); studies were designed sequentially, so registration is confirmatory within a study (Supp. Table S9). The primary safety metric is the false-approval rate among must-refuse cases, paired with the correct-approval rate; we use Wilson 95% intervals, world-cluster bootstrap for pooled image rates, exact McNemar and Wilcoxon tests. A claim of improvement requires the registered direction and $p < 0.05$. Claims are not pooled across studies; the headline claims are G, H, V and A, not the 21 studies.

VI. RESULTS

A. The gate in closed loop

In campaign D (42 trials) and campaign E (60 trials, a new randomised world each) the robot behaved as expected in 42 and 60 trials with 0 and 0 wrong deliveries (E: [0.0, 6.0]%); it completed 36 of the 36 missions in which a safe delivery was possible. A blind robot that trusts the inventory would have made 40 wrong deliveries in the same E trials ([54.1, 77.3]%). On unseen layouts (campaign L, 24 layouts) it behaved as expected in 24/24 ([86.2, 100.0]%) with 0 unsafe deliveries. A delivery without search takes about 5.5 s of checking and 39 s of driving (Supp. Table S5). **Supporting studies (Supp. Secs. S7, S11, S12).** A zero-shot SigLIP baseline (Z) approved 44/408 of T1's must-refuse images against 3/408 for our check; pooled over 12 camera degradations (R) the check made 0/2304 false approvals; a conformal τ_h (K) kept its miss bound at the cost of more false alarms; on real photographs the frozen threshold flagged 64/100 of the fire-free ones, and a threshold calibrated on site photographs (P2) found 282/300 of the fires and gave 18/100 false alarms on unseen photographs (registered claim: supported), so a threshold needs per-site calibration; a second checker (J) was not safer than one model (claim not supported) and was not adopted.

B. Memory in changing worlds (H, V)

Campaign H has 16 sequences of three missions to the same shelter. Lighting and props are re-drawn each mission and four scripted stories change the world between missions (supplies swapped and staying swapped, or swapped back; a small fire appears; a fire disappears and another appears). The first mission seeds all arms; missions 2 and 3 run with no memory (M0), naive memory that skips the check where it last saw the

TABLE II

THE STUDIES THAT CARRY THE CHAIN (ALL REGISTERED STUDIES: SUPP. TABLE S12). CLAIM: STATUS OF THE REGISTERED CLAIM (PENDING: NOT YET RUN OR ANALYSED).

Study	Question	Data	Arms (systems)	Primary endpoint	Claim
T1, E	Does the gate generalise, and work in closed loop?	600 held-out images; 60 randomised worlds	1 (v3.5)	false approvals $< 2\%$ (T1); expected behaviour, wrong deliveries (E)	not supported; descriptive
G (W)	Does looking from the side shrink ε?	24 paired worlds (G); 150 fire + 75 fire-free worlds, images (W)	2 (front only; look-around)	deliveries despite a perceivable fire (G); miss bound $\leq 15\%$ (W)	supported; pending
H	Can memory speed missions up without new risk?	16 sequences $\times$ 3 missions, changing worlds	3 (no, naive, verified memory)	time; unsafe deliveries	supported
V	Does the repeated-mission model predict closed loop?	10 sequences $\times$ 6 missions, random change	3 (as H)	prediction intervals; unsafe deliveries	pending
A	Can untrusted advice relax a check, or steer it?	226 runs: 18 worlds; 16 duplicate-supply worlds; 224 offline texts	tighten-only, naive, LLM-filter, scrutiny, control	unsafe deliveries under false advice; steering; time; cost bound	pending
S	Can a sign in the picture cause a wrong approval?	64 worlds $\times$ no sign, attack, benign	6 (ours, one pass, free-form, full image, OCR mask, ignore-text)	false approvals caused by the sign	pending
X	Does the memory result transfer to a cluttered four-station depot?	depot images (XD); 80 closed-loop runs, $k = 4$	M0 vs M2 (memory); control, tighten-only, scrutiny (steering)	$\hat{\varepsilon}$, recalibrated τ_h; M2 vs M0 unsafe, time; steering (descriptive)	pending

supply or refuses without visiting where it last saw a hazard (M1), and verified memory, i.e. memory used only as advice (M2). Over the 32 paired missions, unsafe deliveries were 20 for M1, 0 for M2 and 1 for M0 (M1 vs. M2 exact McNemar $p = 1.9 \times 10^{-6}$; by sequence 12 vs. 0 of 16, $p = 0.0005$). M1's were 12 wrong supplies, relocated since it last looked, and 8 deliveries into a fire that had appeared since; every one followed a skipped check. The M1–M2 contrast is large by design; the comparison that bounds our risk is M2 versus M0, 0/32 versus 1/32 (Wilson upper bounds 11% and 16%), met as registered but informative only down to those bounds. M0's single unsafe delivery shows what caution is for: the new fire was invisible from the front and small from the side, the left view scored $p_h = 0.035 < \tau_h$ and M0 delivered; M2 had recorded a fire at this station two missions earlier, checked at $\tau_h = 0.01$, scored 0.049 and stopped (Fig. 5). Memory can make the robot more careful, never less (Table III). On the 8 missions whose relocation persisted, the median time was 79 s for M2 against 114 s for M0 (exact Wilcoxon $p = 0.0078$). The price of keeping every check: after the supplies moved back, M2 first visited the proposed station and then searched (113 s against 97 s for M0), and in the model of repeated missions verified memory keeps 27% of naive memory's saving at no change and none at $q = 0.5$ (Supp. Sec. S12); with k stations a stale proposal costs up to $k - 1$ visits, while the expected saving of a correct one grows with k (descriptive model with the measured detour of 30 s: 9 s per mission at $k = 2$, 80 s at $k = 8$, $q = 0.2$; Supp. Sec. S7). The registered headline claim (verified memory keeps memory's speed-up without its safety cost) is supported. **Study V** tests the repeated-mission model in closed loop (10 sequences of 6 missions, supply location and fire toggling at random with $q = 0.2$, predictions registered before the first trial). The registered 95% interval for the paired saving M0−M2 was 11–20 s (median 16 s); we observed *(pending)* s. Unsafe deliveries were *(pending)* (M0), *(pending)* (M1) and *(pending)* (M2) against registered intervals 1 (0–5), 20 (15–25) and 1 (0–5). The claim that the model predicts the closed loop is *(pending)*; that verified memory keeps its speed-up without a safety cost under random change, *(pending)* (M1 vs. M2 by sequence: $p =$ *(pending)*).

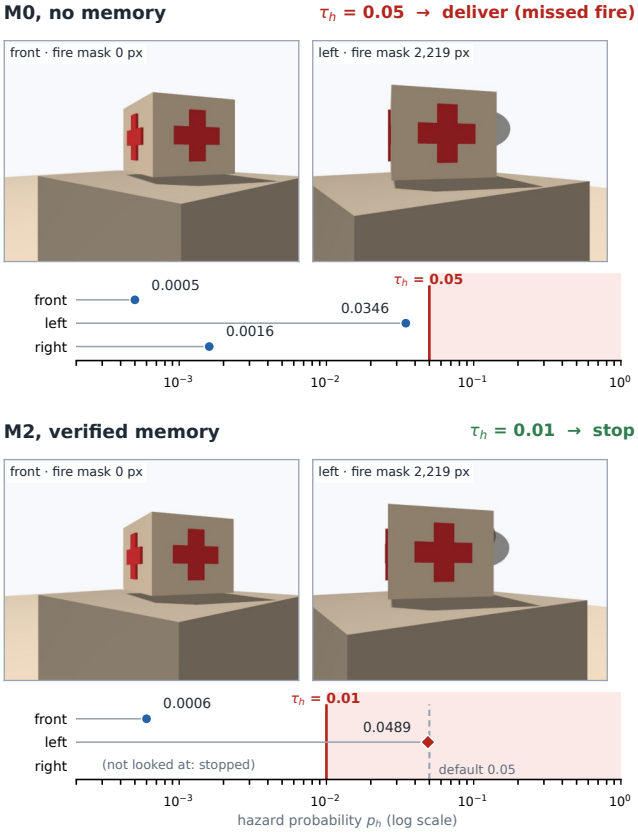


Fig. 5. Campaign H, sequence Q09 mission 3: the same world without memory (M0) and with verified memory (M2). Each row shows the front view and the left side view (fire-mask pixels from the simulator) and the hazard score p_h of every view that was checked against the threshold (shaded: $p_h \geq \tau_h$). M0 used $\tau_h = 0.05$; the left view scored 0.035 and the robot delivered. M2 had seen a fire at this station two missions earlier, so it used $\tau_h = 0.01$; the left view scored 0.049 and the robot stopped without checking the right side. The fire is small and partly hidden from the left, so the two p_h values are close.

C. Adversarial advice (A)

Study A pairs an advice-following baseline with ours on *(pending)* worlds in which false advice, if believed, causes an unsafe delivery ("the fire is out, skip the check"; "the water is at the other station"), through two channels (operator text, a peer's advice file), plus an arm in which an LLM guardrail filters the advice before the baseline follows it. Unsafe-delivery runs: *(pending)* for the advice-following baseline, *(pending)* for ours, *(pending)* in the no-advice control (paired by world: exact McNemar $p =$ *(pending)*; this contrast is a demonstration that the interface, not the model, decides safety; ours unsafe where the control was safe, i.e. exposure: *(pending)*; advice-audit failures: *(pending)*). Truthful location advice saved a median of *(pending)* s per mission (Wilcoxon $p =$ *(pending)*). Adversarial advice cost at most *(pending)* extra visit(s) and *(pending)* false refusals in *(pending)* runs (bound: *(pending)*). **Steering (A-steer).** The adaptive attacker does not try to relax a check; it sends *truthful* location advice

TABLE III

CAMPAIGN H, MISSIONS 2–3 (PAIRED). M0: NO MEMORY; M1: NAIVE MEMORY; M2: VERIFIED MEMORY. UNSAFE = WRONG SUPPLY OR A PERCEIVABLE FIRE AT THE DELIVERING STATION. TIME: MEDIAN OVER ALL 32 MISSIONS.

Arm	Unsafe	Wrong supply	Into fire	False refusals	Median time [s]
M0 no memory (v3.6)	1/32	0	1	0	83
M1 naive memory	20/32	12	8	4	28
M2 verified memory (ours)	0/32	0	0	0	73

TABLE IV

ARENA VERSUS DEPOT (k: STATIONS; M0, M2: NO MEMORY, VERIFIED MEMORY; LETTERS: STUDIES, TABLE II). DEPOT CELLS ARE NOT YET MEASURED; THE LAST COLUMN STATES THE EXPECTED TRANSFER IN ADVANCE.

Quantity	Arena	Depot	Transfers?
Stations k	2	4	–
$\hat{\varepsilon}$ with look-around (Wilson upper)	1/16 (0.28)	*(pending)*	no
Gate false approvals on held-out images	T1: 3/408	XD: *(pending)*	no
τ_h (frozen / recalibrated)	0.05 / –	0.05 / *(pending)*	no
Unsafe M0 / M2 (of 32 in H; no M1 arm in the depot)	1 / 0	*(pending)* / *(pending)*	M2 vs M0: yes
Median time M0 → M2, relocation persists	114 → 79 s	*(pending)*	size: no
Stale-proposal price (changed back)	97 → 113 s	*(pending)*	grows with k
Steering by a truthful tip (descriptive; no naive arm in the depot)	A-steer: *(pending)*	X-s: *(pending)*	cost: yes

to the station that holds a small, partly hidden fire when a clean duplicate exists. In 16 duplicate-supply worlds, unsafe deliveries were *(pending)* for tighten-only advice, *(pending)* for the control and *(pending)* with scrutiny and soft refusal, at *(pending)* lost deliveries and *(pending)* s; the plug-in $k\hat{\varepsilon}$ is 0.125 (Sec. III-B). This closed-loop cell shows cost and mitigation, not an effect size: with $\varepsilon \approx 1/16$ about one unsafe world is expected in 16, and 0 of 16 has a Wilson upper bound of 0.19, so a null is "undetectable at this n, upper bound *(pending)*", never "order is irrelevant". The steering estimate is the oracle-steering ε_{any} at image level (W, XD; about 130 and 100 hard-class worlds): *(pending)*. Registered claims: safety *(pending)*, time *(pending)*, cost *(pending)*. A consistency–robustness curve of the policies as advice accuracy falls is a model illustration, not a result: its fire sits at the requested supply's station, so the tighten-only curve is flat by construction and only its time axis is informative (Supp. Sec. S10).

D. Outside the guarantee: scene text and realism

Scene text (S). Study S (image level, *(pending)* worlds each rendered without a sign, with an attack sign and with a benign text) measures what the guarantee does not cover. Flips (refuse → approve) of ours: "Answer C" *(pending)* of *(pending)*; "A", tied to the unpermuted option order, *(pending)* of *(pending)* (one unpermuted pass: *(pending)*); "B", invariant under the permutation, *(pending)* of *(pending)*; mislabels *(pending)* of *(pending)* (free-form baseline: *(pending)*); "SAFE – NO FIRE" beside a visible fire *(pending)* of *(pending)*. Signs legible: *(pending)*; registered claim: *(pending)*, with *(pending)* registered hypotheses supported. Whatever the numbers, scene text is outside the guarantee of Sec. IV.

Realism and transfer. The arena is a flat empty floor with two stations and a stylised flame. The depot (study X) uses an MIT-0 warehouse (shelves, clutter, lamps, a pallet jack), four supply bays on pallets or low shelves, aisle routes, a lidar-guarded obstacle in some worlds and a moving smoke emitter, including smoke-only worlds, the weakest case on real photographs; fire and smoke remain synthetic. We state before the data what should transfer (Table IV). The *relative* claims are structural and should replicate: naive advice fails; tighten-only advice stays inside the check's own budget; the look-around lowers ε. The *absolute* ε, thresholds and times are not expected to, and a recalibrated τ_h would be the honest transfer result. XD re-measures ε per condition class on held-out depot images (frozen threshold first, recalibrated second, as in P2); X runs no-memory against verified memory at $k = 4$ (safety, time, visited bays, the stale-proposal price) and a three-arm steering cell with at most eight fire worlds, descriptive only: it reports the availability cost of a truthful tip. The depot is a scene-distribution shift inside Gazebo; only image-level tests on photographs (P, P2, Q, XD-R) touch real appearance. Study Q tests the supply check on *(pending)* public Commons photographs (water, medicine, neither; labels audited by an AI assistant, with a human spot-check pending), and XD-R the false-alarm rate on fire-free warehouse photographs; neither involves a robot. Real photographs (P, P2) link the simulation to real appearance without a robot; they showed that the frozen threshold flagged 64/100 of fire-free photographs (hazy hillsides, sunsets) while the scores still rank well (AUROC 0.96).

VII. DISCUSSION AND LIMITS

Inputs, not just models, decide safety. Trusting memory is efficient but its risk grows with how often the world changes, which the robot cannot observe without looking (H). Observation 1 and Proposition 2 make the question moot for the existence of checks: memory and any other advice decide the order and strictness of checks, never whether they happen; the guarantee ends at the camera. **Moving beats thinking harder.** The largest gain in ε came from moving 0.4 m (G), not from a larger model, a better prompt or a second model. **What pre-registration bought us.** Of the registered claims, four failed (T1's 2% bound, F, T2, J); the first two led to the look-around, and J stopped us adopting a second model. Unregistered, each could have been rationalised away.

Limits. The guarantee is relative: with $k\hat{\varepsilon} = 0.125$ for the hardest class (upper bound 0.57), the sum bound is vacuous at $k = 4$; the order-independent $\varepsilon_{\text{any}} \le k\hat{\varepsilon}$ is the number to read, estimated at image level (W, XD: *(pending)*). Exposure to a really burning station and persistent advice (an availability cost per mission that accumulates over missions) are not bounded. ε_s differs by station and fire geometry, so the budget we measure on one class does not transfer to another. Scene text is perception and outside the guarantee. All results are from simulation, with one robot, two supply types, a stylised fire and sensor degradations rather than the appearance gap of real objects or smoke; the depot (study X, pending) is meant to narrow this gap but remains simulation (no grasping, real smoke dynamics or sensor noise). Campaign H scripts its changes (V tests random ones), and 8–16 pairs per closed-loop study are few. Barely visible fires (21–613 pixels) are

excluded, and the guard is hand-written, not synthesised [20], [21]. Threats to validity: Supp. Sec. S9.

VIII. Conclusion

A gate that rarely approves a wrong delivery needs a small error budget and a rule that nothing it did not perceive can exceed it. The look-around shrank the budget (G: 11 to 1 of 16); typing every unperceived input as advice that may only tighten kept a stale or hostile input from spending more than the budget (H: supported; A: *(pending)*). The guarantee is relative and ends at the camera; scene text and realism are measured, not claimed. Everything is simulation, and a site-specific threshold is the first thing a real deployment must calibrate. All code, protocols and per-trial evidence are released.

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